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II. MATHEMATICAL MODELING

THE FORWARD PROBLEM FOR SURFACE WAVE DISPERSION IN LAYERED MEDIA

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UDC 519.612

We examine a new method for the calculation of the traveling wave characteristic in a layered medium. The seismic impedance tensor method is improved by introducing the notion of a potential function and new iterative relationships are obtained for isotropic layered media. Comparison of the dispersion equation roots between the generalized reflection-transmission coefficient method (GR/TC) and the seismic impedance tensor method (SIT) has detected some roots that do not exist in SIT. To check the accuracy of the roots in both methods, we have compared the experimental results between the classical transition matrix method (Thomson-Haskell) and our new method (SIT) for the same layered medium model. The experimental results show that the roots obtained by the impedance method coincide with the classical transition matrix method. This also shows that some roots obtained from the dispersion equation in GR/TC cannot be accurately determined.

Keywords: traveling wave characteristic, impedance tensor method, dispersion curve, phase velocity

Introduction

Surface wave studies are a new method in geophysical sounding. Features of this method include high energy and high signal-to-noise ratios. In recent years, the method has been widely used in small-depth geophysical sounding (above the Moho surface, about 30–40 km deep), in deep seismological studies, and in many other scientific areas.

The forward problem of seismic surface waves is mostly studied by calculating the dispersion of the Rayleigh surface wave. The classical method for a piecewise-constant plane-layered medium is the Thomson-Haskell transition matrix method [3]. In practice, numerical calculations involve loss of high-frequency accuracy. To overcome this difficulty, a number of new methods have been developed [4–11]. Chen Xiaofei has proposed the generalized reflection-transmission coefficient method (GR/TC) for calculating the dispersion of a plane wave [12–15]. This method finds a correct root of the dispersion equation, i.e., it provides a "root search" process. The complexity of the root search process increases as more layers are considered.

The present author has derived a new dispersion equation by introducing the seismic impedance tensor [1]. A system of differential equations has been developed for the solution of the dispersion equation. Solving this system by a Runge-Kutta method of the fourth order, we have obtained the dispersion curve for the normal-mode Rayleigh wave. However, the computation time and accuracy of this method for the solution of the differential system are sensitive to the choice of the grid increment (if the increment is large, loss of accuracy is observed; if the increment is small, computations slow down). The present article derives a new iterative relationship for the seismic impedance tensor that overcomes this weakness.

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The Seismic Impedance Tensor Method

Consider a layered medium with the boundary $z = z_n$, $n \in [0, n]$, where $z_0 = 0$ is the Earth's surface. Inside each layer, $z \in [z_{n-1}, z_n]$, $n \in [1, N]$. The seismic parameters λ_n, μ_n, ρ_n are assumed constant. A traveling wave propagates along the OX axis in this medium. The traveling-wave displacement vector is specified in the form

$$\vec{U}(x, y, z) = \bar{u}(z) e^{i\gamma x + i\omega t} \tag{1}$$

where ω is the frequency of the seismic field, γ is the traveling-wave propagation constant (characteristic), and $\bar{u}(z)$ is the traveling-wave amplitude. Inside the layer, $\bar{u}(z)$ satisfies the Lame equation with constant coefficients:

$$(\lambda_n + 2\mu_n) \operatorname{grad} \operatorname{div} \vec{U} - \mu_n \operatorname{rot} \operatorname{rot} \vec{U} + \omega^2 \rho_n \vec{U} = 0, \quad z \in [z_{n-1}, z_n]. \tag{2}$$

At the layer boundaries we have continuity conditions on the displacement $\bar{U}(z)$ and the stress $\{\sigma_{ij}\}$, $i, j \in [x, y, z]$, where

$$\sigma_{ij} = \lambda \delta_{ij} \operatorname{div} \bar{U}(z) + \mu \left(\frac{\partial \bar{U}(z)}{\partial x_j} + \frac{\partial \bar{U}(z)}{\partial x_i} \right), \quad \{x_i\} = \{x, y, z\}, \quad i \in [1, 3]. \tag{3}$$

The seismic impedance tensor of a surface wave links the stresses σ_{xz}, σ_{zz} with the displacements u_x, u_z by linear relationships:

$$\begin{cases} \sigma_{xz} = Z_{xx} u_x + Z_{xz} u_z, \\ \sigma_{zz} = Z_{zx} u_x + Z_{zz} u_z, \end{cases} \tag{4}$$

where

$$\hat{Z} = \begin{vmatrix} Z_{xx} & Z_{xz} \\ Z_{zx} & Z_{zz} \end{vmatrix} \tag{5}$$

$\hat{Z}$ is the seismic impedance tensor of rank 2. The impedance tensor is continuous on the discontinuities of λ, μ, and ρ because the stresses and the displacements are continuous.

Defining the impedance tensor at $z = 0$, i.e., $\hat{Z}(z = 0) = \hat{Z}^0$, we obtain from the boundary conditions for the stress at $z = 0$

$$\begin{aligned} \sigma_{xz}(z = 0) &= Z^0_{xx} u_x(z = 0) + Z^0_{xz} u_z(z = 0) = 0, \\ \sigma_{zz}(z = 0) &= Z^0_{zx} u_x(z = 0) + Z^0_{zz} u_z(z = 0) = 0. \end{aligned} \tag{6}$$

A surface traveling wave exists (i.e., nonzero $u_x(0)$ and $u_z(0)$ exist) only if

$$\det \hat{Z}^0 = Z_{xx}^0 Z_{zz}^0 - Z_{xz}^0 Z_{zx}^0 = 0. \tag{7}$$

This is the dispersion equation that determines the traveling-wave propagation constant γ as a function of the frequency ω and the parameters of the layered medium.

Equation (7) is the forward problem for the calculation of the dispersion curve of surface waves. The impedance $\hat{Z}^0(\overline{\lambda},\overline{\mu},\overline{\rho},\overline{h},\omega,\gamma)$ has been obtained from the system of equations in the form

$$\begin{cases} \dfrac{dZ_{xx}}{dz} = i\lambda\mu\gamma\left(Z_{xz}-Z_{zx}\right)-\left(\lambda+2\mu\right)Z_{xx}^2-\mu Z_{xz}Z_{zx}+4\mu^2\gamma^2\left(\lambda+\mu\right)-\omega^2\rho\mu\left(\lambda+2\mu\right), \\ \dfrac{dZ_{xz}}{dz} = -\left(\left(\lambda+2\mu\right)i\mu\gamma Z_{xx}-i\mu\lambda\gamma Z_{zz}-\left(\lambda+2\mu\right)Z_{xx}Z_{xz}-\mu Z_{xz}Z_{zz}\right), \\ \dfrac{dZ_{zx}}{dz} = -\left(\left(\lambda+2\mu\right)i\mu\gamma Z_{xx}-i\gamma\mu\lambda Z_{zz}+\left(\lambda+2\mu\right)Z_{zx}Z_{xx}+\mu Z_{zx}Z_{zz}\right), \\ \dfrac{dZ_{zz}}{dz} = -\left(\left(\lambda+2\mu\right)i\mu\gamma Z_{xz}-\left(\lambda+2\mu\right)i\mu\gamma Z_{zx}+\left(\lambda+2\mu\right)Z_{xz}Z_{zx}+\mu Z_{zz}^2+\omega^2\rho\mu\left(\lambda+2\mu\right)\right). \end{cases} \tag{8}$$

To solve the system, we have to specify the initial value of the impedance tensor at $z = z_N = H$ (the half-space surface). Given $Z\ (z=H)=\hat{Z}^H$ it is easy to solve the Cauchy problem numerically for $\hat{Z}(z)$:

$$\begin{cases} Z_{xx}^{(H)} = -\dfrac{\mu^{(H)}\eta_2{}^{(H)}\left(\eta_1^{2(H)}-\gamma^2\right)}{\eta_1{}^{(H)}\eta_2{}^{(H)}-\gamma^2}, \\ Z_{xz}^{(H)} = -\dfrac{i\mu^{(H)}\gamma\left(\eta_1^{2(H)}-2\eta_1{}^{(H)}\eta_2{}^{(H)}+\gamma^2\right)}{\eta_1{}^{(H)}\eta_2{}^{(H)}-\gamma^2}, \\ Z_{zx}^{(H)} = \left(\lambda^{(H)}+2\mu^{(H)}\right)\dfrac{i\gamma\left(\eta_1{}^{(H)}\eta_2{}^{(H)}-\eta_2^{2(H)}\right)}{\eta_1{}^{(H)}\eta_2{}^{(H)}-\gamma^2}+i\gamma\lambda^{(H)}, \\ Z_{zz}^{(H)} = \left(\lambda^{(H)}+2\mu^{(H)}\right)\dfrac{\eta_1{}^{(H)}\left(\eta_2^{2(H)}-\gamma^2\right)}{\eta_1{}^{(H)}\eta_2{}^{(H)}-\gamma^2}. \end{cases} \tag{9}$$

Experiments show that the choice of the increment in the process of numerical solution directly influences both the computation speed and the accuracy of the forward algorithm. We accordingly introduce a potential function that produces a new iterative relationship for the seismic impedance tensor in a layered medium.

Iterative Relationships Derived from the Seismic Impedance Tensor for Isotropic Layered Media

We introduce the functions $u_s(z)$ and $u_p(z)$ for the solenoidal and potential parts of the displacement field, respectively. Then in a Rayleigh surface traveling wave we have by (1)

$$\begin{aligned} u_x &= -u_s'(z) + i\gamma u_p(z), \\ u_z &= i\gamma u_s(z) + u_p'(z). \end{aligned} \tag{10}$$

Here $u_s(z)$ and $u_p(z)$ are solutions of the following equations:

1. $u_s'' - \eta_s^2 u_s = 0$, where $\eta_s = \sqrt{\gamma^2 - k_s^2}$, $k_s = \dfrac{\omega}{v_s}$, $\operatorname{Re}\eta_s > 0$, $v_s = \sqrt{\dfrac{\mu}{\rho}}$ is the transverse wave velocity;

2. $u_p'' - \eta_p^2 u_p = 0$, where $\eta_p = \sqrt{\gamma^2 - k_p^2}$, $k_p = \dfrac{\omega}{v_p}$, $\operatorname{Re}\eta_p > 0$, $v_p = \sqrt{\dfrac{\lambda + 2\mu}{\rho}}$ is the longitudinal wave velocity.

In both cases, $\gamma > k_s > k_p$.

Consider the field in layer $z \in [z_{n-1}, z_n]$ with the constants λ_n, μ_n, ρ_n, where $n \in [1, N]$, $z_0 = 0$, $z_N = -\infty$, $h = z_n - z_{n-1}$ is the layer thickness. In layer n, $u_s(z)$ and $u_p(z)$ can be represented in the form

$$\begin{aligned} u_s(z) &= \overset{(n)}{A}_s e^{\overset{(n)}{\eta}_s(z - z_n)} + \overset{(n)}{B}_s e^{-\overset{(n)}{\eta}_s(z - z_n)}, \\ u_p(z) &= \overset{(n)}{A}_p e^{\overset{(n)}{\eta}_p(z - z_n)} + \overset{(n)}{B}_p e^{-\overset{(n)}{\eta}_p(z - z_n)}. \end{aligned} \tag{11}$$

Then in this layer the displacement-field components are

$$u_x(z) = i\gamma\left(\overset{(n)}{A}_p e^{\overset{(n)}{\eta}_p(z - z_n)} + \overset{(n)}{B}_p e^{-\overset{(n)}{\eta}_p(z - z_n)} \right) - \overset{(n)}{\eta}_s \left(\overset{(n)}{A}_s e^{\overset{(n)}{\eta}_s(z - z_n)} - \overset{(n)}{B}_s e^{-\overset{(n)}{\eta}_s(z - z_n)} \right), \tag{12}$$

$$u_z(z) = i\gamma\left(\overset{(n)}{A}_s e^{\overset{(n)}{\eta}_s(z - z_n)} + \overset{(n)}{B}_s e^{-\overset{(n)}{\eta}_s(z - z_n)} \right) + \overset{(n)}{\eta}_p \left(\overset{(n)}{A}_p e^{\overset{(n)}{\eta}_p(z - z_n)} - \overset{(n)}{B}_p e^{-\overset{(n)}{\eta}_p(z - z_n)} \right). \tag{13}$$

The stress tensor (3), allowing for (4), has the following two components in layer n:

$$\sigma_{xz} = \mu\left(i\gamma u_z(z) + u_x'(z) \right) = Z_{xx} u_x + Z_{xz} u_z, \tag{14}$$

$$\sigma_{zz} = (\lambda + 2\mu) u_z'(z) + \lambda i\gamma u_x(z) = Z_{zx} u_x + Z_{zz} u_z. \tag{15}$$

Substituting (12), (13) in the left-hand side of (14)–(15), we find

$$\sigma_{xz} = \mu\left(-\left(\gamma^2 - \overset{(n)}{\eta}_s^2 \right)\left(\overset{(n)}{A}_s e^{\overset{(n)}{\eta}_s (z-z_n)} - \overset{(n)}{B}_s e^{-\overset{(n)}{\eta}_s (z-z_n)} \right) \right.$$

$$\left. + 2i\gamma \overset{(n)}{\eta}_p \left(\overset{(n)}{A}_p e^{\overset{(n)}{\eta}_p (z-z_n)} - \overset{(n)}{B}_p e^{-\overset{(n)}{\eta}_p (z-z_n)} \right) \right), \tag{16}$$

$$\sigma_{zz} = 2i\gamma\mu \overset{(n)}{\eta}_s \left(\overset{(n)}{A}_s e^{\overset{(n)}{\eta}_s (z-z_n)} - \overset{(n)}{B}_s e^{-\overset{(n)}{\eta}_s (z-z_n)} \right)$$

$$+ \left((\lambda + 2\mu) \overset{(n)}{\eta}_p^2 - \lambda\gamma^2 \right)\left(\overset{(n)}{A}_p e^{\overset{(n)}{\eta}_p (z-z_n)} - \overset{(n)}{B}_p e^{-\overset{(n)}{\eta}_p (z-z_n)} \right). \tag{17}$$

The parts $\left((\lambda + 2\mu) \overset{(n)}{\eta}_p^2 - \lambda\gamma^2 \right)$ in Eq. (17) satisfy the relationship

$$(\lambda + 2\mu) \eta_p^2 - \lambda\gamma^2 = (\lambda + 2\mu)\left(\gamma^2 - k_p^2 \right) - \lambda\gamma^2$$

$$= 2\mu\gamma^2 - \rho\omega^2 = \mu\left(\gamma^2 + \eta_s^2 \right).$$

The stress (17) thus can be rewritten as

$$\sigma_{zz} = 2i\gamma\mu \overset{(n)}{\eta}_s \left(\overset{(n)}{A}_s e^{\overset{(n)}{\eta}_s (z-z_n)} - \overset{(n)}{B}_s e^{-\overset{(n)}{\eta}_s (z-z_n)} \right)$$

$$+ \mu\left(\gamma^2 + \overset{(n)}{\eta}_s^2 \right)\left(\overset{(n)}{A}_p e^{\overset{(n)}{\eta}_p (z-z_n)} - \overset{(n)}{B}_p e^{-\overset{(n)}{\eta}_p (z-z_n)} \right). \tag{18}$$

Then for layer n at $z = z_n$ (the bottom boundary) the displacement and the stress can be represented in the form

$$u_x(z_n) = i\gamma\left(\overset{(n)}{A}_p + \overset{(n)}{B}_p \right) - \overset{(n)}{\eta}_s \left(\overset{(n)}{A}_s - \overset{(n)}{B}_s \right),$$

$$u_z(z_n) = i\gamma\left(\overset{(n)}{A}_s + \overset{(n)}{B}_s \right) + \overset{(n)}{\eta}_p \left(\overset{(n)}{A}_p - \overset{(n)}{B}_p \right), \tag{19}$$

$$\sigma_{xz}(z_n) = \mu\left(-\left(\gamma^2 - \overset{(n)}{\eta_s^2}\right)\left(\overset{(n)}{A_s} - \overset{(n)}{B_s}\right) + 2i\gamma\,\overset{(n)}{\eta_p}\left(\overset{(n)}{A_p} - \overset{(n)}{B_p}\right)\right),$$

$$\sigma_{zz}(z_n) = 2i\gamma\mu\,\overset{(n)}{\eta_s}\left(\overset{(n)}{A_s} - \overset{(n)}{B_s}\right) + \mu\left(\gamma^2 + \overset{(n)}{\eta_s^2}\right)\left(\overset{(n)}{A_p} - \overset{(n)}{B_p}\right). \tag{20}$$

At $z = z_{n-1}$ (the top boundary),

$$u_x(z_{n-1}) = i\gamma\left(\overset{(n)}{A_p}\, e^{\overset{(n)}{\eta_p} h} + \overset{(n)}{B_p}\, e^{-\overset{(n)}{\eta_p} h}\right) - \overset{(n)}{\eta_s}\left(\overset{(n)}{A_s}\, e^{\overset{(n)}{\eta_s} h} - \overset{(n)}{B_s}\, e^{-\overset{(n)}{\eta_s} h}\right),$$

$$u_z(z_{n-1}) = i\gamma\left(\overset{(n)}{A_s}\, e^{\overset{(n)}{\eta_s} h} + \overset{(n)}{B_s}\, e^{-\overset{(n)}{\eta_s} h}\right) + \overset{(n)}{\eta_p}\left(\overset{(n)}{A_p}\, e^{\overset{(n)}{\eta_p} h} - \overset{(n)}{B_p}\, e^{-\overset{(n)}{\eta_p} h}\right), \tag{21}$$

$$\sigma_{xz}(z_{n-1}) = \mu\left(-\left(\gamma^2 - \overset{(n)}{\eta_s^2}\right)\left(\overset{(n)}{A_s}\, e^{\overset{(n)}{\eta_s} h} - \overset{(n)}{B_s}\, e^{-\overset{(n)}{\eta_s} h}\right) + 2i\gamma\,\overset{(n)}{\eta_p}\left(\overset{(n)}{A_p}\, e^{\overset{(n)}{\eta_p} h} - \overset{(n)}{B_p}\, e^{-\overset{(n)}{\eta_p} h}\right)\right),$$

$$\sigma_{zz}(z_{n-1}) = 2i\gamma\mu\,\overset{(n)}{\eta_s}\left(\overset{(n)}{A_s}\, e^{\overset{(n)}{\eta_s} h} - \overset{(n)}{B_s}\, e^{-\overset{(n)}{\eta_s} h}\right) + \mu\left(\gamma^2 + \overset{(n)}{\eta_s^2}\right)\left(\overset{(n)}{A_p}\, e^{\overset{(n)}{\eta_p} h} - \overset{(n)}{B_p}\, e^{-\overset{(n)}{\eta_p} h}\right). \tag{22}$$

Given the seismic impedance tensor $\hat{Z}(z_n)$, we now rewrite (20) and (21) as the system of equations

$$\begin{cases} \mu\left(-\left(\gamma^2 - \overset{(n)}{\eta_s^2}\right)\left(\overset{(n)}{A_s} - \overset{(n)}{B_s}\right) + 2i\gamma\,\overset{(n)}{\eta_p}\left(\overset{(n)}{A_p} - \overset{(n)}{B_p}\right)\right) - Z_{xx}u_x(z_n) + Z_{xz}u_z(z_n) = 0, \\ 2i\gamma\mu\,\overset{(n)}{\eta_s}\left(\overset{(n)}{A_s} - \overset{(n)}{B_s}\right) + \mu\left(\gamma^2 + \overset{(n)}{\eta_s^2}\right)\left(\overset{(n)}{A_p} - \overset{(n)}{B_p}\right) - Z_{zx}u_x(z_n) + Z_{zz}u_z(z_n) = 0, \\ u_x(z_{n-1}) - i\gamma\left(\overset{(n)}{A_p}\, e^{\overset{(n)}{\eta_p} h} + \overset{(n)}{B_p}\, e^{-\overset{(n)}{\eta_p} h}\right) - \overset{(n)}{\eta_s}\left(\overset{(n)}{A_s}\, e^{\overset{(n)}{\eta_s} h} - \overset{(n)}{B_s}\, e^{-\overset{(n)}{\eta_s} h}\right) = 0, \\ u_z(z_{n-1}) - i\gamma\left(\overset{(n)}{A_s}\, e^{\overset{(n)}{\eta_s} h} + \overset{(n)}{B_s}\, e^{-\overset{(n)}{\eta_s} h}\right) + \overset{(n)}{\eta_p}\left(\overset{(n)}{A_p}\, e^{\overset{(n)}{\eta_p} h} - \overset{(n)}{B_p}\, e^{-\overset{(n)}{\eta_p} h}\right) = 0. \end{cases} \tag{23}$$

Solving this system, we can represent the parameters $\overset{(n)}{A_s}$, $\overset{(n)}{B_s}$, $\overset{(n)}{A_p}$, $\overset{(n)}{B_p}$ in the form

$$\begin{cases} \overset{(n)}{A_s} = K_1\left(\hat{Z}(z_n)\right)u_x(z_{n-1}) + K_2\left(\hat{Z}(z_n)\right)u_z(z_{n-1}), \\ \overset{(n)}{B_s} = K_3\left(\hat{Z}(z_n)\right)u_x(z_{n-1}) + K_4\left(\hat{Z}(z_n)\right)u_z(z_{n-1}), \\ \overset{(n)}{A_p} = K_5\left(\hat{Z}(z_n)\right)u_x(z_{n-1}) + K_6\left(\hat{Z}(z_n)\right)u_z(z_{n-1}), \\ \overset{(n)}{B_p} = K_7\left(\hat{Z}(z_n)\right)u_x(z_{n-1}) + K_8\left(\hat{Z}(z_n)\right)u_z(z_{n-1}). \end{cases} \tag{24}$$

Substituting (24) in (22), we obtain

$$\begin{cases} \sigma_{xz}(z_{n-1}) = Q_1\left(\hat{Z}(z_n)\right)u_x(z_{n-1}) + Q_2\left(\hat{Z}(z_n)\right)u_z(z_{n-1}), \\ \sigma_{zz}(z_{n-1}) = Q_3\left(\hat{Z}(z_n)\right)u_x(z_{n-1}) + Q_4\left(\hat{Z}(z_n)\right)u_z(z_{n-1}), \end{cases} \tag{25}$$

where K_1, K_2, K_3, K_4, K_5, K_6, K_7, K_8, Q_1, Q_2, Q_3, Q_4 are parameters dependent on the tensor $\hat{Z}(z_n)$ and the constants λ_n, μ_n, ρ_n, h_n.

We have derived iterative relationships for the seismic impedance tensor in layer n:

$$\hat{Z}(z_{n-1}) = \begin{vmatrix} Q_1\left(\hat{Z}(z_n)\right) & Q_2\left(\hat{Z}(z_n)\right) \\ Q_3\left(\hat{Z}(z_n)\right) & Q_4\left(\hat{Z}(z_n)\right) \end{vmatrix}. \tag{26}$$

Note that this iterative relationship fundamentally obviates the differential equation (8) given in the literature [1]. Specifying the layered-medium model and using the value of $\hat{Z}_H$ for (9) at $z = H$, we can easily solve the system of equations and find the iterative relationship for $\hat{Z}_0$.

Experimental Results

We specify a simple experimental model of a layered medium (Table 1). Instead of the constants ρ, Vp, Vs, we can numerically calculate the Lame coefficients λ, μ for each layer. Substituting all the constants in (7), (23)–(26), we obtain all the characteristics γ and phase velocities c that depend on the corresponding period (Fig. 1). Seismic data processing usually focuses on the normal-mode traveling wave of frequency less than 1 Hz. In our numerical experiments, we accordingly determine the dispersion curve of the normal-mode traveling wave in the frequency interval 10–200 sec.

The new iterative tensor method produces results that closely match the traditional impedance tensor method, with errors of around 1% (Fig. 1). The new iterative impedance tensor relationship is thus suitable for studying the traveling wave dispersion, uniquely separating the normal-mode traveling wave. This establishes the efficiency of the proposed method.

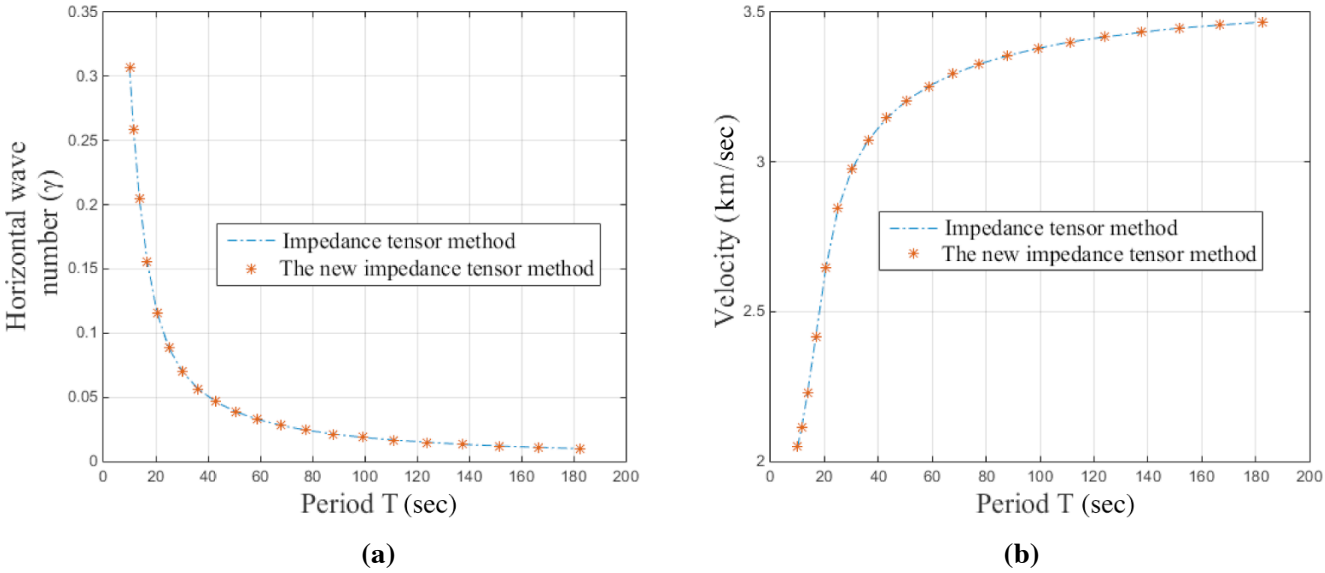


Fig. 1. a) Comparison of the traveling-wave characteristic γ obtained by the new iterative impedance tensor method and the traditional impedance tensor method; b) Comparison of the phase velocities obtained by the new iterative impedance tensor method and the traditional impedance tensor method.

Table 1
The Coefficients of the Layered Medium in Model 1

Layer	Depth (km)	ρ (g/cm^3)	Vp (km/sec)	Vs (km/sec)
1	10	2.4	3.5	2.1
2	20	2.5	4	2.7
3	∞	3	6	4

Comments: Vp is the longitudinal wave velocity, Vs is the transverse wave velocity.

Analysis of the Rayleigh Surface Wave Multimode

Having defined a new solution technique for the dispersion equation and specified a layered-medium model (Table 1), we proceeded with a detailed analysis and comparison of the roots with those obtained by the GR/TC method (Fig. 2). We have identified the following difficulties:

1. At low frequencies, GR/TC produces roots that are smaller than the root representing the phase velocity of the normal-mode surface wave. These roots are without physical meaning (Figs. 2a, 2b, 2c, 2d). The first root obtained by the impedance method is always the root representing the phase velocity of the normal-mode surface wave.

2. With the exception of the cases described in point 1 above, GR/TC produces roots that are not detected by the impedance tensor method.

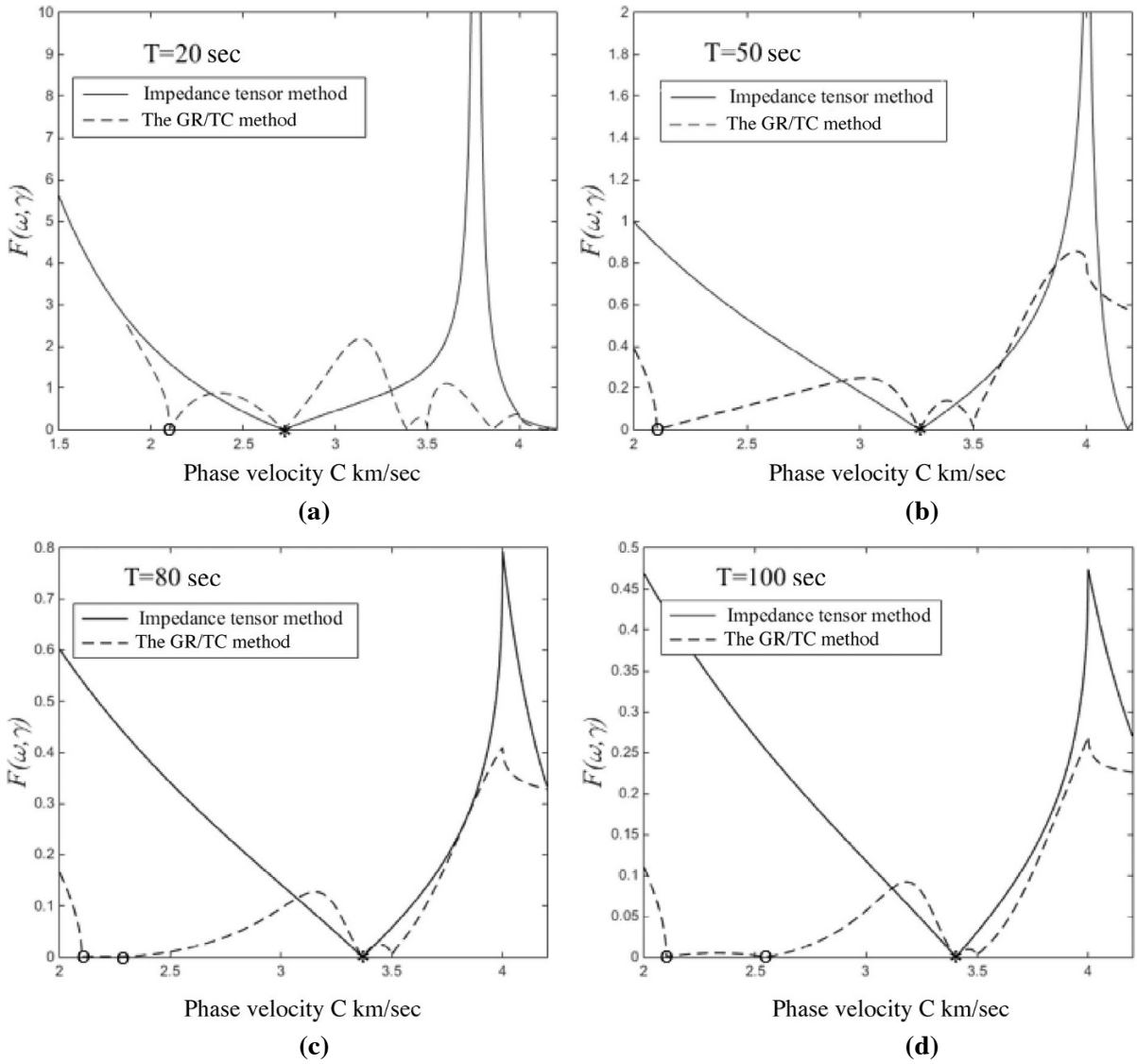


Fig. 2. Low-frequency results for $T = 20, 50, 80, 100$. The asterisk * denotes the root representing the phase velocity of the normal-mode surface wave and the circle ○ denotes roots below the phase velocity of the normal-mode surface wave.

To check the accuracy of the roots produced by the two methods, we have compared the experimental results for the classical Thomson-Haskell matrix transition method with those for our new seismic impedance method using the particular model of the layered medium (Table 1). The experimental results show that the roots obtained by the impedance method coincide with the classical matrix transition method (Fig. 3). This also shows that some roots of the dispersion equation obtained by GR/TC cannot be accurately determined.

CONCLUSION

We thus reach the following conclusions.

1. Although GR/TC resolves the difficulty with the loss of accuracy in the Thomson-Haskell classical matrix transition algorithm, the roots obtained from the dispersion equation include some “false” (inaccurate) roots

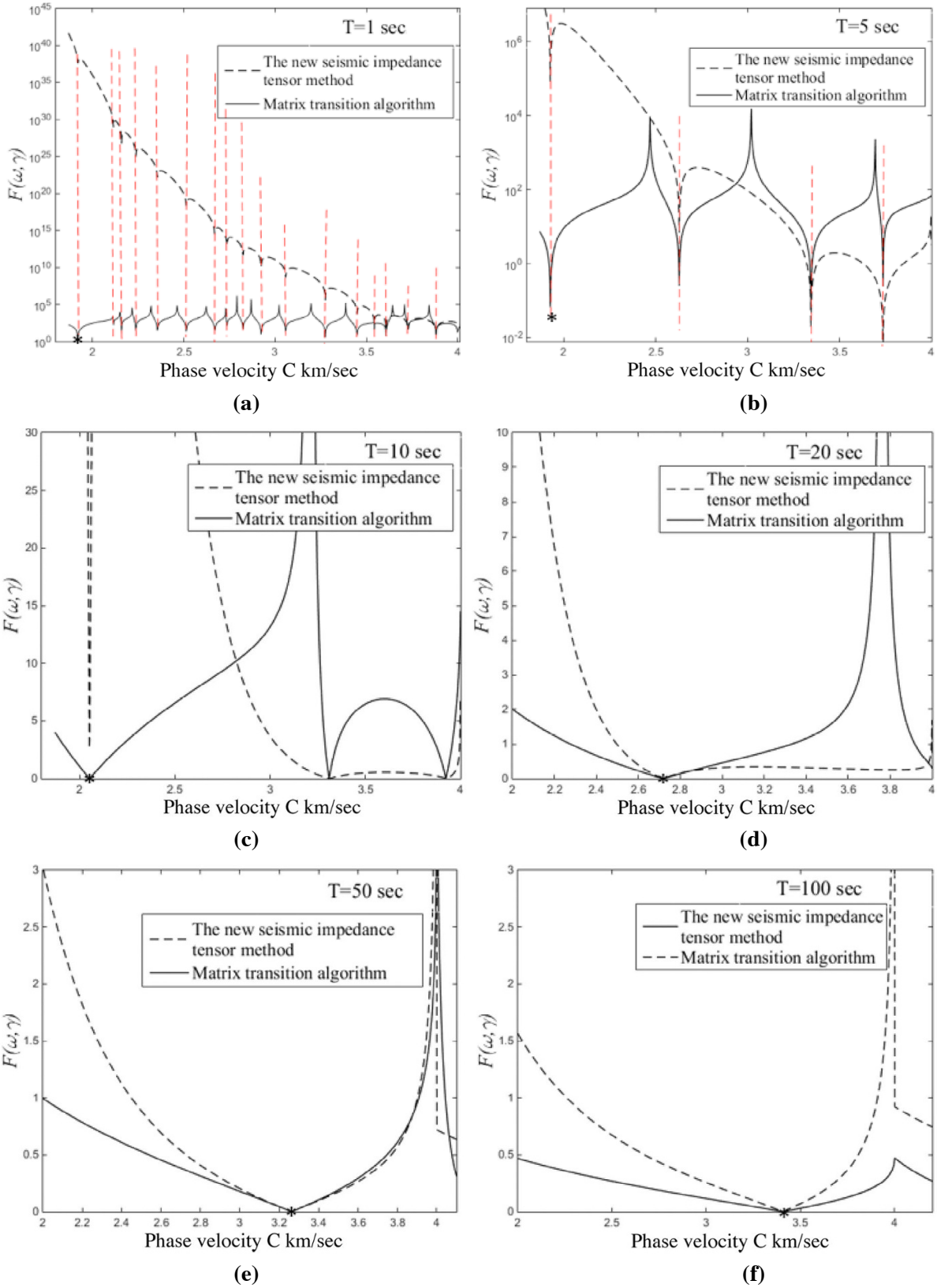


Fig. 3. Results for different frequencies T = 1, 5, 10, 20, 50, 100 sec (panels a, b, c, d, e, f, respectively). The asterisk * is the root representing the phase velocity of the normal-mode surface wave.

that are physically meaningless. These redundant roots affect the computational speed and accuracy. When computing the dispersion curve for each surface-wave mode, we have to determine which of the roots are physically meaningful; this involves a "root search" process. As the number of layers in the model is increased, the "root search" process becomes more complicated and time consuming. The seismic impedance tensor method does not require such a search. Each root obtained from the dispersion equation has its true physical meaning, and the first root always gives the phase velocity of the normal-mode surface wave.

2. The seismic impedance tensor method also resolves the difficulty with high-frequency loss of accuracy, which is observed in the classical matrix transition method. Furthermore, the root obtained by the impedance tensor method is the phase velocity of each surface-wave mode in the high-frequency range.

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